

Project for the “Workshop: Software tools for machine learning (a.a. 2025/26)”

Due date: May 11, 2026

The assigned dataset has been downloaded from the Kaggle website ([LINK](#)) and adapted for the given assignment. A description of the corresponding features is given at the previous link. Each student is assigned a different version of the previous dataset (info on Ariel).

As a consequence, your dataset might also miss have different features and items with respect to the one of your colleagues.

The students have to address the following points:

1. Data cleaning. Each feature with more than 30% of empty cells has to be discarded. In the other case, with less than 30% of empty cells, the corresponding items have to be removed from the dataset.
2. Data encoding. Some of the categorical variables might be informative or not for the following steps. As a consequence you could either decide to encode or simply to drop those features. Discuss your choice before finalizing the preprocessed dataset.
3. Use an appropriate machine learning algorithm to infer the degree of satisfaction of the travelers ('satisfied' or 'neutral or dissatisfied').
4. Verify if some of the features are correlated with others.
5. Determine which are the least and the most relevant features with respect to the degree of satisfaction (point 2).

Considerations:

1. Each project has to be submitted as a single Python notebook and save according to the rule “name_surname_ID.ipynb”. Each choice and result must be explained in the notebook using as markdown text.
2. You are free to use any type of algorithm in the project. If you are willing to use an algorithm that has not been presented along the course, you must explain the reason of your choice and the details of the methodology you adopted (together with online references)
3. Although the use of IA tools is not forbidden, consider that these instruments often generate clumsy code, that is not always bug free. Therefore, each student must carefully check its own code before submitting the project.

Any project that is not fully documented or has any of the issues reported before will necessarily require a separate oral session before its approval.